Lesson Plan

Middle School Science

The Importance of Air Flow

Pedagogical Rationale

Big Ideas

Environmental Sciences – Air quality

Living and non-living things respond to their environment

Content Knowledge

- Sense and respond to the environment
- Phases of matter - gasses

Essential Questions

(a) How does air quality affect living and non-living (biotic and abiotic) things?
(b) How do living and non-living things detect, respond, and adapt to changes in air quality?
(c) How can we identify and measure the different substances in the air?
(d) How do we change the mixture of the air around us?
(e) What are the effects of changes in air quality?
(f) Why is it important to analyze air quality?

Learning Objectives:

Students will be able to:

- Describe how air is comprised of different gasses.
- Predict changes in air quality when a variable is changed.
- Explore how we can measure changes of the components of the air around us.
- Design air quality experiments for the RADbox to measure.
- Examine the results of their experiments to explain differences in air quality.
- Assess and justify how air quality can be improved.
- Imagine processes or initiatives that could improve air quality.
- Connect their creative thinking and application of skills by suggesting scenarios that will change the composition of the air.
- Connect to their critical thinking and application of knowledge by observing the changes in air quality recorded by the RADbox and exploring the implications for living and/or non-living things caused by those changes.

Curricular Competencies:[1]
Students will have opportunities to show they can:

Question and predict
- Demonstrate curiosity about the natural world
- Observe objects and events in familiar contexts
- Identify questions about familiar objects and events that can be investigated scientifically
- Make predictions based on prior knowledge

Plan and conduct an inquiry
- Suggest ways to plan and conduct an inquiry to find answers to their questions
- Consider ethical responsibilities when deciding how to conduct an experiment
- Safely use appropriate tools to make observations and measurements, using formal measurements and digital technology as appropriate
- Make observations about living and non-living things in the local environment
- Collect simple data

Process and analyze data and information
- Experience and interpret the local environment
- Identify First Peoples perspectives and knowledge as sources of information
- Sort and classify data and information using drawings or provided tables
- Use tables, simple graphs, or other formats to represent data and show simple patterns and trends
- Compare results with predictions, suggesting possible reasons for findings

Evaluate
- Make simple inferences based on their results and prior knowledge
- Reflect on whether an investigation was a fair test
- Demonstrate an understanding and appreciation of evidence
- Identify some simple environmental implications of their and others' actions

Apply knowledge and innovate
- Contribute to care for self, others, school, and neighborhood through individual or collaborative approaches
- Cooperatively design projects
- Transfer and apply learning to new situations
- Generate and introduce new or refined ideas when problem solving

Communicate and collaborate
- Represent and communicate ideas and findings in a variety of ways, such as diagrams and simple reports, using digital technologies as appropriate
- Express and reflect on personal or shared experiences of place.

1. *Science 4*. BC's Course Curriculum, 2016.
https://curriculum.gov.bc.ca/curriculum/science/4/core

Indigenous Perspectives:[2]

This lesson supports the following First Peoples principles of learning:

- Learning ultimately supports the well-being of the self, the family, the community, the land, the spirits, and the ancestors.
- Learning is holistic, reflexive, reflective, experiential, and relational (focused on connectedness, on reciprocal relationships and a sense of place)
- Learning involves recognizing the consequences of one's actions.

These principles are brought to life through the power of learning by doing and through participation in an authentic, high-level experiment. Students employ their critical thinking skills through investigating the results of the experiments and extrapolating the implications of changes in air quality for individuals and communities. This activity invites discussion surrounding care for biotic and abiotic things - ourselves, our communities, and our environment. For example, if the air measures high in humidity, what may happen to cultural treasures made of organic materials?

2. *The First Peoples Principles of Learning.* First Nations Education Steering Committee, 2007. https://www.fnesc.ca/first-peoples-principles-of-learning/

Inclusive and Equitable Practice:

In this lesson, inclusive and equitable practices are integrated to create a welcoming and engaging environment for all students. The RADbox has been designed with the intention of allowing any student to address any hypothesis they may have about the air quality around them and the mobility of the device itself allows students to explore a wide range of environments and cultures beyond the classroom. Furthermore, the lesson planned in particular involves student feedback in a group setting where all students are included and encouraged to participate. Lastly, the students were able to engage in peer-to-peer group work to share these perspectives further. The mode of communication of ideas is not prescriptive and can, therefore, be adapted to meet all students' learning needs as required. Throughout, differentiation and support can be incorporated as student needs require, ensuring every student can participate and learn effectively.

Assessment Design

Formative	Summative	Metacognitive
Throughout this lesson, formative assessments are woven into the activities to gauge student understanding and provide timely feedback. The development of preparatory knowledge is the first opportunity for students to display their understanding. By engaging in discussion around typical air quality sensor use in real-world applications, specifically measuring the factors examined, students are able to demonstrate an understanding of why and how air quality is typically measured. During the data collection period, students are encouraged to discuss how the air quality factors might be changing around	Learning can be assessed against the aforementioned learning objectives in a variety of formats at the teacher's discretion. It could be easily be achieved through a threefold observation of student engagement with the material: ● First, the exploration of measuring the different components that make up the air around us both in real-life applications following the initial presentation as well as in an experiment setting with the use of the RADbox. ● Second, the quality of hypotheses regarding how different factors affect air quality during data collection.	The participatory nature of this experiment encourages students to reflect on their learning throughout the lesson. The iterative nature of the tasks requires students to explore and refine their ideas and understanding of the material continuously. The interactive nature of the lesson encourages active engagement and deepens understanding, fosters self-awareness and empowers students to become more effective scientists.

them as well as discussing the graphical representation of the experiment as data is collected. Through this discussion, students are able to establish hypotheses as to why air quality might change in response to the air quality factors changing as well as seeing the rate with which this change may occur. From here, students can grow understanding around the differing changes from different effects on the air quality factors. Lastly, by breaking into peer-to-peer group work to establish further experiments for evaluating air quality factors, the students are able to demonstrate the background knowledge they have developed throughout the lesson as well as explore hypotheses from the data collected. These ongoing assessments ensure that students receive feedback throughout the lesson.	● Lastly, the demonstration of critical thinking skills regarding the changes in air quality and the extent to which conclusions can be drawn against hypotheses and establish further, more complex hypotheses.	

<table>
<tr><th colspan="4">Suggested Lesson Structure</th></tr>
<tr><td>RESOURCES</td><td colspan="3">● RADbox
● Laptop computer (not a ChromeBook)
● A method of displaying RADbox results to the students (e.g. projector screen)
● Samples of air quality changing substances to wave in front of the RADbox sensor: e.g. rubbing alcohol, air freshener, a candle, match, vinegar, glue, bunsen burner, natural gas tap, fan
● Large pieces of paper for recording group work ideas or dry-erase boards
● Marker pens
● Graphic organizer (a template of a potential worksheet is included)</td></tr>
<tr><td>PREPARATION</td><td colspan="3">● Become acquainted with the RADbox, how it operates, and how its graphs are displayed to students in real-time.
● Download the necessary RADbox software to a laptop computer to be used during the lesson.
● Gather samples and stationery supplies needed for the experiment (listed above).
● Prior to the lesson, record up to 30 minutes (360 scans) of air quality data before students are present with the windows and doors closed. Note how many people are present during this collection. Name the raw data .csv file “Baseline”.
● Prepare the environmental conditions for the classroom. In order for a room to be ideal for this experiment, there should be either a window or door which is able to be opened on opposite walls of the room.</td></tr>
<tr><td>PREPARATORY KNOWLEDGE</td><td colspan="3">Prior to this lesson, it is useful if students have an understanding of :
● air being a mixture
● convection currents
● phase transitions
● scientific instruments (data collection and management)
● non-living things sensitivity to air quality (wood, fabrics)</td></tr>
<tr><td>SPECIALIST VOCABULARY</td><td colspan="3">Prior to or during this lesson, students will need to understand the following specialist vocabulary:
● volatile organic compounds (VOCs)
● carbon dioxide (CO_2)
● nitrogen oxide species (NO_X)
● mixture</td></tr>
<tr><td></td><td>Time</td><td colspan="2">Teaching and Learning Activities</td></tr>
<tr><td rowspan="2">ENGAGE</td><td rowspan="2">5-10 mins</td><td>Teacher</td><td>Students</td></tr>
<tr><td>Begin the lesson with windows closed. Allow all students and teachers to enter the room and let the air in the room settle for approximately 5-10 minutes.

During this waiting period, do a warm-up activity to introduce the lesson to students, activate prior knowledge, engage them in the lesson, encourage them to make predictions, and to focus them on the task at hand (e.g. complete the first two sections of a Know/Want-to Know/ Learned (KWL) chart)</td><td>Engage in teacher-led discussion regarding real-world applications of air quality data as well as begin to think critically around what might cause the air quality to change quickly. Complete KWL Chart.</td></tr>
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		Introduce the Real-time Air Data Box (RADbox) to students as a device which records carbon dioxide, volatile organic compounds, nitrogen oxide species, temperature, and humidity. Students should be familiar with these terms as well as simple ways these factors may affect air quality before the experiment begins. Explain that the RADbox is able to measure air quality in real time while providing instant graphical feedback. If necessary, demonstrate the software being used to the classroom without the presence of additional chemicals. Have the students break into peer-to-peer groups (3-4 students) and have them guess how different substances in general may affect air quality. What might those substances be? How might they get into the air in the first place? This task can be achieved through drawing expected graphs on pieces of paper depicting air quality change. Compare and consolidate the ideas generated by each group as a class.	Collaborate with peers to generate ideas about how different substances will affect air quality. Share ideas with the class and actively listen to others' ideas. Listen to the explanations and ask questions to help clarify understanding of the apparatus.
EXPLORE then EXPLAIN	40 mins	Present the substances (e.g. candle flame, aerosol spray of choice, rubbing alcohol) that have been chosen to affect air quality to the students. Ask what hypotheses can be made following the group discussion around graphing. Note these hypotheses somewhere all the students are able to see, such as a central dry-erase board. Place students in groups (3-4 students) and ask them to sketch expected graphs of air quality over time showing how air quality might change for each substance. Use dry-erase boards or large pieces of paper to record results. Compare student predictions. Discuss why some graphs are different from others as well as why some graphs may be similar between groups. (5 minutes) What (if any) safety precautions would scientists need to heed during this experiment?	Observe the qualities of each substance. Contribute to whole-class discussion by making hypotheses about the effect of each substance. Work collaboratively to compare what factors are believed to change air quality for each substance. Sketch your group's predictions for how the RADbox graph will appear when each substance is introduced. Pay attention to how quickly the change might happen in the graph and how large the change is. Engage in class discussion. Draw hypotheses, as a class, as to how the introduced substances might affect air quality during data collection.

		Note the time, record the amount of students present, and run the RADbox application. Run for 30 minutes (360 scans). Distribute the RADbox worksheet. During the first 10 minutes of data collection, introduce the substances which are expected to change the air quality to the environment surrounding the RADbox. This period specifically observes how the air quality in the room is affected by the substances introduced. Compare the RADbox data to the students' predictions and hypotheses. 10 minutes (120 scans) later, open one of the windows/doors. Have students note observations for any noticeable changes in the room in the supplied worksheet and discuss expected changes in the data. Discuss if there is anything in the environment outside the room that might affect the air quality directly. Encourage students to explore both positive and negative effects that may be expected. After another 10 minutes (240 scans total), open the second window/door opposite the first. In the case where there is no second window/door available, a fan can be used to generate airflow. Note observations for any noticeable changes in the room in the supplied worksheet and discuss expected changes in the data. Discuss if there is a noticeable change in the airflow in the room. Encourage students to discuss signs that show the airflow has changed such as temperature or intensity. Finally, after 5 more minutes (300 scans total). A fan may be added to one of the windows/doors, should equipment be available. The fan may blow in or out in either window/door and students are encouraged to discuss any differences between the two options. For a final time, note observations for any noticeable changes in the room in the supplied worksheet and discuss expected changes in the data.	Observe the data collected during the RADbox experiment and engage in the resulting class discussion. Record what is seen in the first ten minutes in the first section of the supplied worksheet. Observe in the second part of the worksheet how the data collected changes after the window is opened. Make note not only of how the data shows these changes but what else in the room reflects that the air quality has changed. Does the room feel hotter or colder? Is there a perceptible difference in humidity? Continue to make observations in the worksheet as the experiment continues with the second window/door opening. Listen to the presentation and observe the qualities of each substance. Contribute to whole-class discussion by making hypotheses about the effect of each substance. Continue to make observations in the worksheet as the experiment continues with the fan being used. Listen to the presentation and observe the qualities of each substance. Contribute to whole-class discussion by making hypotheses about the effect of each substance.

		Once all scans are complete, observe how the created figures in the RADbox program represent the data. Are there any clear trends? Are there any changes in the graph that are aligned with expectations? Are there any unexpected changes? These questions all may serve to either drive direct discussion or serve as inspiration for research questions which students may be asked to design on their own. In general: ● Volatile organic compounds are commonly found in household chemicals such as rubbing alcohol, whiteboard cleaner, vinegar, perfume, and air fresheners. Presence of these factors are expected to increase the values of volatile organic compounds in the experiment. Increased airflow to the experiment environment should lower these values once the sources have been removed from the experiment. ● Nitrogen oxide species are produced by transportation pollution, oil and gas pollution, and fossil fuel/electric power plants. If these factors are present in the environment outside the classroom, then the values of nitrogen oxide species are expected to rise when exposed to airflow from outside during experimentation. ● Carbon dioxide is commonly produced from breathing, combustion/fire, and pollution. Presence of these factors are expected to increase the values of carbon dioxide in the experiment. Increased air flow to the experiment environment should lower these values once the sources have been removed from the experiment unless the values outside are typically higher than the values inside given factors such as pollution. ● Temperature and humidity are expected to change as the airflow to the classroom is increased, especially if the classroom is climate controlled. Furthermore, steaming water or combustion/fire are expected to increase temperature and humidity individually. Finally, compare the raw data .csv generated during experimentation to the raw data .csv collected as “Baseline”. What are the differences in the results? What might be expected to cause these differences? What can these differences tell us about air quality during class time?	

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		What might be expected to cause these differences? What can these differences tell us about air quality during class time?	what you can do to affect air quality directly and how you might be able to have the largest effect on the air quality. Complete the third section of the worksheet. Compare the graphs created by the teacher to the graphs you had created as a hypothesis. What is similar and what is different? Use this comparison to complete the fourth section of the worksheet.
	There is a natural break point here to split this lesson over two sessions if needed.		
ELABORATE and EXTEND	10+ mins	Allow the students to break off into groups (3-4 students). Distribute a large piece of paper and markers to each group. Have students compare completed worksheets. Encourage comparison of factors which might change air quality and the completed hand drawn graphs. Should the teacher see fit, these suggestions can then be paired with the RADbox for further experimentation. Further testing of substances could be performed by the teacher or the students, at the teacher's discretion.	In groups (3-4 students), compare the completed worksheets. What points do people have in common? Which points are unique? Specifically focus discussion on the fourth section of the worksheet, the biggest changes to air quality. As a group, record which factors you have decided would have the biggest changes on air quality with the supplied dry-erase boards/paper and markers. Choose one person in the group to represent the group in class discussion. Suggest further substances for the RADbox to test. Predict what may happen to air quality and compare with actual results.
EVALUATE	10+ mins	Have the students record the factors which they have decided as a group might change air quality onto the large pieces of paper. Use dry-erase boards or large pieces of paper to record results. After a 5 minute group discussion period, meet as a class and compare the factors which might affect air quality between groups. What would improve/worsen air quality? What actions can we take or habits can we practice to improve air quality? What other ethical/practical/environmental ideas has the class pursued? Once the class has regrouped, have the group representative share the ideas you have developed as a group on how the air quality can be changed with the class.	Collaborate with peers to recap, refine, and expand on ideas about how different substances will affect air quality. Be prepared to defend your ideas: What would be the effect on living and non-living things in the environment? What actions can we take to mitigate worsening air quality? What actions can we take or habits can we practice to improve air quality? What other ethical/practical/environmental ideas has the class pursued? Participate in the class discussion that shares each group's ideas. Complete KWL chart.

		Have students complete the “Learned” section of their KWL chart or use another method of reflective practice to draw students’ attention to their cognitive growth.	